Network Science

Lecture 05: Social Context and Link Formation

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Speaker notes

Lectures 03 and 04 treated the graph as a structure of anonymous nodes and unlabeled edges — the only question was "is the edge there?". This lecture adds context: nodes get *attributes* (age, track, beliefs), edges get *timestamps*, and the graph gets *foci* (shared social settings). The central tension shifts from the **computational** gap of L03–L04 (what we want to compute is NP-hard) to a **causal** gap: the mechanism behind a network pattern — selection vs. socialization — cannot be identified from a single snapshot. The fix is not a better algorithm but richer data: longitudinal observations and explicit social contexts.

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From Structure to Context

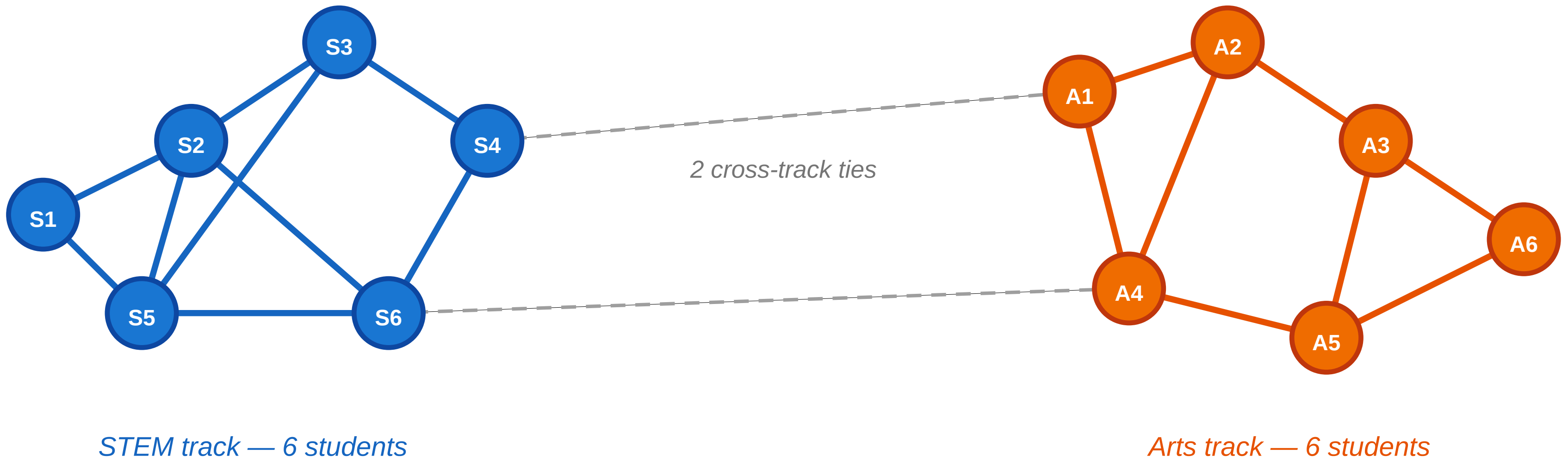
In Lectures 03 and 04 we classified edges (strong vs. weak) and groups (communities). The graph was colorless — every node was interchangeable. Today we ask: what happens when nodes carry *attributes* and edges carry *reasons*?

Consider a small classroom.

A Motivating Scenario

A small classroom: friendships and attributes

circles = students · color = track (blue = STEM, orange = Arts) · lines = friendships



Questions we'd like to answer from this picture

① Is the within-track clustering more than we'd expect by chance?
② Did students choose friends from the same track, or did they become more alike after befriending?
③ Could shared classes (foci) explain the clustering without any personal preference? ④ Would mild preferences suffice to create this pattern?

> Twelve students, split equally between STEM (blue) and Arts (orange). Most friendships are within-track; only two cross-track ties exist. The clustering is visible — but why is it there?

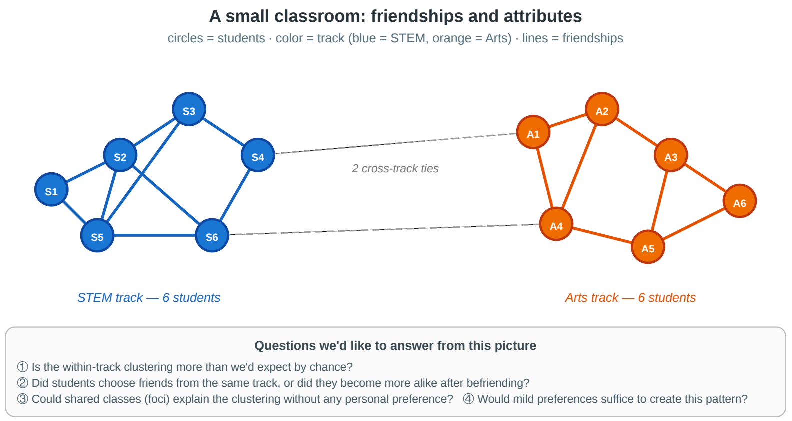
Speaker notes

This figure is the anchor for the whole lecture. Use it to provoke the four questions on the next slide before any definition arrives. By the end of the session students should be able to articulate four different explanations for the visible pattern and say which data would distinguish them.

Four Questions from One Picture

1. **Measurement.** Is the within-track clustering more than we would expect by chance — or could it happen under random mixing?
2. **Mechanism.** Did students choose friends from the same track (**selection**), or did friends influence each other to pick the same track (**socialization**)?
3. **Context.** Could shared classes or labs (**foci**) explain the clustering without any personal preference for similarity?
4. **Dynamics.** If every student has only a mild preference for same-track friends, could that suffice to produce this sharp segregation?

Every concept in today's lecture answers one of these.



Speaker notes

Each question maps to a module: (1) homophily and its formal baseline, (2) selection vs. socialization and the causal identification problem, (3) affiliation networks and focal closure, (4) online link formation and (5) Schelling segregation as a model for how mild local preferences generate sharp global patterns. Keep referring back to the classroom figure throughout.

A Different Kind of Gap

In L03–L04 the gap between theory and measurement was **computational**: MaxSTC and modularity maximization are NP-hard, so we used polynomial-time heuristics.

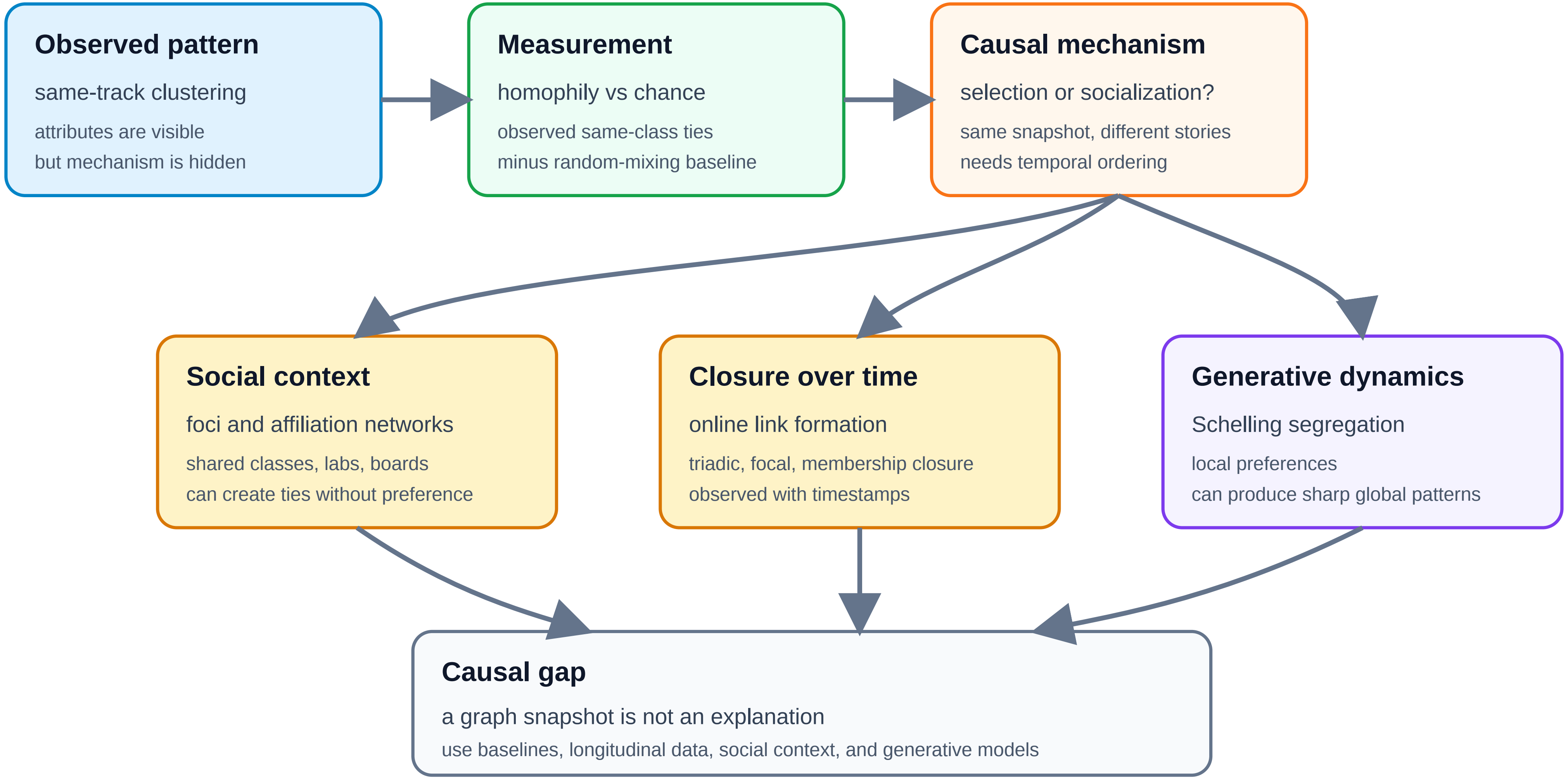
In L05 the gap is **causal**: even with infinite compute, a cross-sectional snapshot cannot tell us *why* a tie formed. The fix is not a better algorithm — it is richer data.

Question	What we want to know	What a snapshot shows	What we need
Homophily?	Is tie formation biased toward similarity?	Within-group edge fraction	Random-mixing baseline
Selection or socialization?	Which came first — similarity or the tie?	Both present simultaneously	Longitudinal data
Foci or preference?	Did shared context cause the tie?	Shared memberships and ties	Affiliation network model
Local → global?	Can mild preferences produce sharp segregation?	A segregated pattern	A generative model (Schelling)

Speaker notes

This slide is the L05 analogue of L03's "theory vs. measurement" and L04's roadmap table. The new axis is causal identification rather than computational tractability. The rest of the lecture fills in the four rows.

Argument Map



Speaker notes

This is the L05 analogue of the L04 synthesis diagram. The lecture starts from an observed pattern: same-track clustering in a friendship graph. It first asks whether the pattern exceeds a random-mixing baseline, then asks which mechanism could have produced it: selection, socialization, shared foci, closure over time, or local preferences amplified by Schelling dynamics. The bottom row is the central lesson: a graph snapshot is evidence, not an explanation; causal interpretation requires baselines, temporal data, social context, and generative models.

Takeaway

> L03–L04 asked "what structure is there?" and hit a computational wall. L05 asks "why is the structure there?" and hits a causal wall. Network analysis requires both algorithmic tools and causal reasoning about the mechanisms behind the observed graph.

Speaker notes

This frames the shift cleanly: from computation to causation. Refer back to this framing when each module introduces its specific gap.

Homophily

> **Guiding Question:** Do people form ties because they are similar — and how would we know?

Speaker notes

The key point is that homophily is a *statistical* pattern requiring a counterfactual baseline. "Many within-group ties" is not evidence of homophily unless we know how many ties we would expect under random mixing. Students should leave this module knowing that declaring homophily requires a baseline, not just a visual impression.

What Does Homophily Look Like?

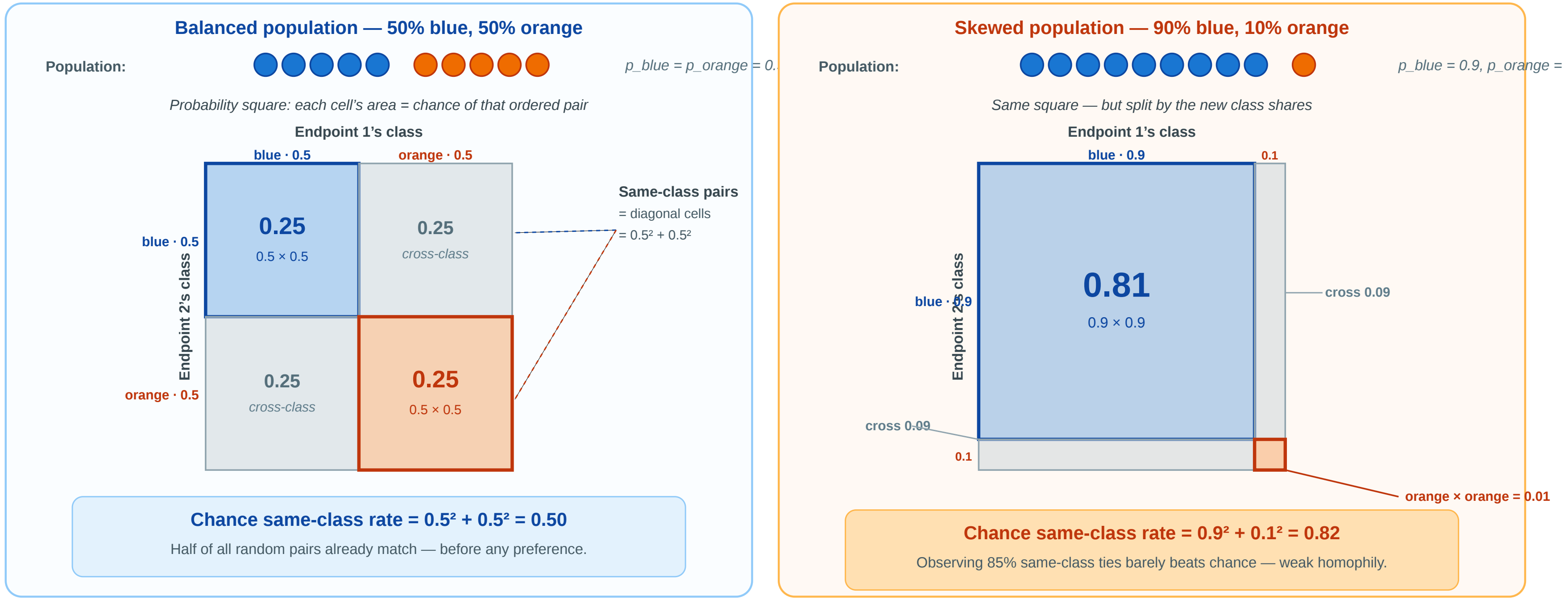
Homophily = tie formation is biased toward similarity. Within-group edges (colored) dominate; cross-group edges (gray) are rare. The pattern looks clear-cut — but a strong within-group share could also arise mechanically when one class is much larger than the other. The next slide makes that worry precise.

Speaker notes

This is the visual hook for the whole homophily module. Students see the pattern first — clustered, segregated, easy to spot — then in the next slide they discover that "easy to spot" is not the same as "more than chance." Use this slide to land the everyday meaning of homophily ("birds of a feather flock together") before any formula appears.

Homophily — What Counts as "More Than Chance"?

Before calling that pattern *homophily*, we have to ask: **how many same-class ties would we see if people paired up purely at random?** Only the *excess* over that chance rate is evidence of a preference for similarity.



Each cell's area is the chance of that ordered pair; the **colored diagonal cells** are the same-class outcomes. In the skewed population (right), 82% of random pairs already match — so an observed 85% within-group rate is almost noise. *Next slide:* we name this chance rate H_{base}.

Speaker notes

This slide opens with the *question* — what would chance alone produce? — not with a formula. The probability-square visualisation lets students *see* the answer: each cell is an ordered pair of endpoints, its area is the joint probability of that pair, and the diagonal cells are the same-class outcomes. The skewed 90/10 panel is the punchline: chance alone explains 82% within-group ties, so any honest claim of homophily must beat that rate. Only on the next slide do we attach the name H_{base} and the formula $\sum p_i^2$ — by that point students already understand what the formula is doing geometrically: summing the areas of the diagonal cells.

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Homophily — Formal Definition

Homophily. Edge formation is biased toward similarity in a node attribute x. Formally, for a categorical attribute with class shares $p_1, \ldots, p_k$:

- **Random-mixing baseline** (area of the diagonal cells in the probability square):

$$H_{\text{base}} = \sum_{i=1}^{k} p_i^2$$

- **Observed homophily:** the fraction of actual ties that connect same-class nodes, H_{obs}.
- **Homophily index:** how much the observed same-class rate exceeds the random baseline, normalized:

$$r = \frac{H_{\text{obs}} - H_{\text{base}}}{1 - H_{\text{base}}}$$

$r = 0$: random mixing. $r = 1$: perfect segregation. $r < 0$: heterophily (preference for *different* attributes).

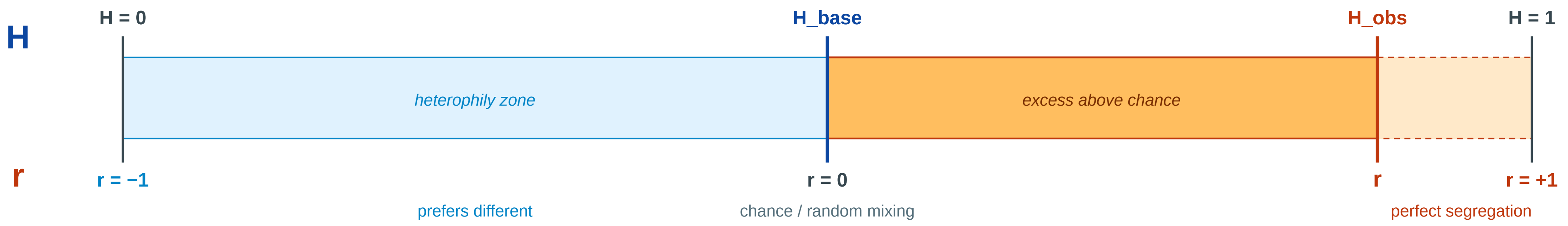
Speaker notes

With the probability-square intuition from the previous slide, the formula $H_{\text{base}} = \sum p_i^2$ is just the area of the diagonal cells. The homophily index r is a variant of Newman's assortativity coefficient for categorical attributes (Newman, 2003); it is the standard way to quantify homophily in empirical network studies. The normalization by $(1 - H_{\text{base}})$ matters precisely because of the 90/10 case shown in the previous figure: reporting "85% same-group" without the baseline would overstate the evidence for homophily.

Homophily — Three Numbers, One Story

Measure	What it tells you	Depends on
H_{base}	**The chance rate.** What random pairing alone would produce. Large when one class dominates the population.	Class shares only
H_{obs}	**What the data actually shows.** Count the within-class edges and divide by the total.	Observed edges only
r	**How far the network has moved from chance toward perfect segregation**, on a 0–1 scale. $r = 0$: chance. $r = 1$: perfect segregation. $r < 0$: cross-class preference (heterophily).	Both

one bar, two readings: H along the top, r along the bottom



r = (H_obs − H_base) / (1 − H_base)

r < 0 when H_obs falls below the chance baseline H_base

Speaker notes

This slide gives each of the three measures a one-sentence plain-English meaning and shows r as a *fractional position* on the line between chance and perfect segregation. The visual is meant to dispel a common confusion: r is **not** $H_{\text{obs}} - H_{\text{base}}$ (the raw excess), but the *fraction of available room* the network has used. The "available room" is whatever lies above the chance baseline; in a balanced population the room is large (0.50), in a skewed 90/10 population the room is small (0.18). That denominator is what makes r comparable across populations with different class shares.

Homophily Baseline Trap

The **same observed within-group share** can mean very different things depending on the population composition. The table below holds $H_{\text{obs}} = 0.85$ fixed and varies the class shares:

Population shares	H_{base}	Room above chance	r	What you can honestly say
50% / 50%	0.50	0.50	0.70	Strong excess — 70% of the way to perfect segregation
70% / 30%	0.58	0.42	0.64	Substantial — 64% of available room used
90% / 10%	0.82	0.18	0.17	Barely above chance — almost nothing to explain
99% / 1%	0.98	0.02	< 0	*Below* chance — actually heterophilic!

Literature example. McPherson, Smith-Lovin & Cook distinguish **baseline homophily** from **inbreeding homophily** (McPherson et al. 2001). A school can show many same-age friendships simply because classes, lunch periods, and activities are age-graded; a workplace can show many same-occupation ties because people in the same job category share meetings and tasks. The real homophily question is not "are many ties same-type?", but "are there **more** same-type ties than the opportunity structure already predicts?"

> Never interpret "X% of ties are within-group" without asking: **X% compared to what baseline?** The baseline depends on the population, not on what feels intuitively high.

Speaker notes

This is a useful didactic pause because students often read a high within-group percentage as direct evidence of strong preference. The expanded table shows that the same 0.85 within-group rate maps to $r = 0.70$ in a balanced population but to $r = 0.17$ — or even *heterophily* — in skewed populations. A concrete teaching example: a school that is 80% one ethnic group and 20% another will *mechanically* produce ~68% same-ethnicity friendships even under fully random mixing, before any preference. Reporting "82% same-group" out of context would overstate the evidence by a wide margin. This is precisely why the McPherson, Smith-Lovin & Cook (2001) review of homophily research demands a baseline.

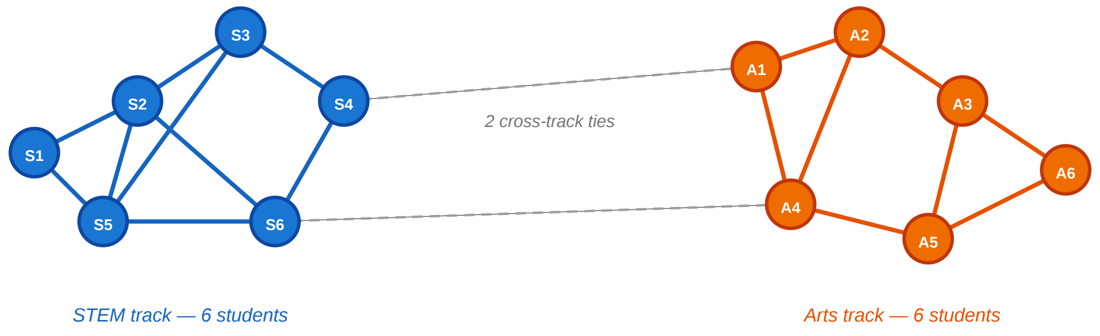
Back to the Classroom

Both tracks have 6 students, so $p_{\text{STEM}} = p_{\text{Arts}} = 0.5$.

- $H_{\text{base}} = 0.5^2 + 0.5^2 = 0.50$
- 17 within-track edges of 19 total: $H_{\text{obs}} = 17/19 \approx 0.89$
- $r = (0.89 - 0.50)/(1 - 0.50) = 0.78$

A small classroom: friendships and attributes

circles = students · color = track (blue = STEM, orange = Arts) · lines = friendships



Questions we'd like to answer from this picture

① Is the within-track clustering more than we'd expect by chance?
② Did students choose friends from the same track, or did they become more alike after befriending?
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Speaker notes

Walk through the arithmetic side-by-side with the figure: point at the within-track clusters as you read off H_{obs}, then point at the two dashed cross-track ties as the remainder. The punchline — "strong homophily, but the mechanism is unknown" — is the bridge to Module 2. The same $r = 0.78$ would arise under pure selection, pure socialization, or pure focal closure. Distinguishing them requires more than a cross-sectional snapshot.

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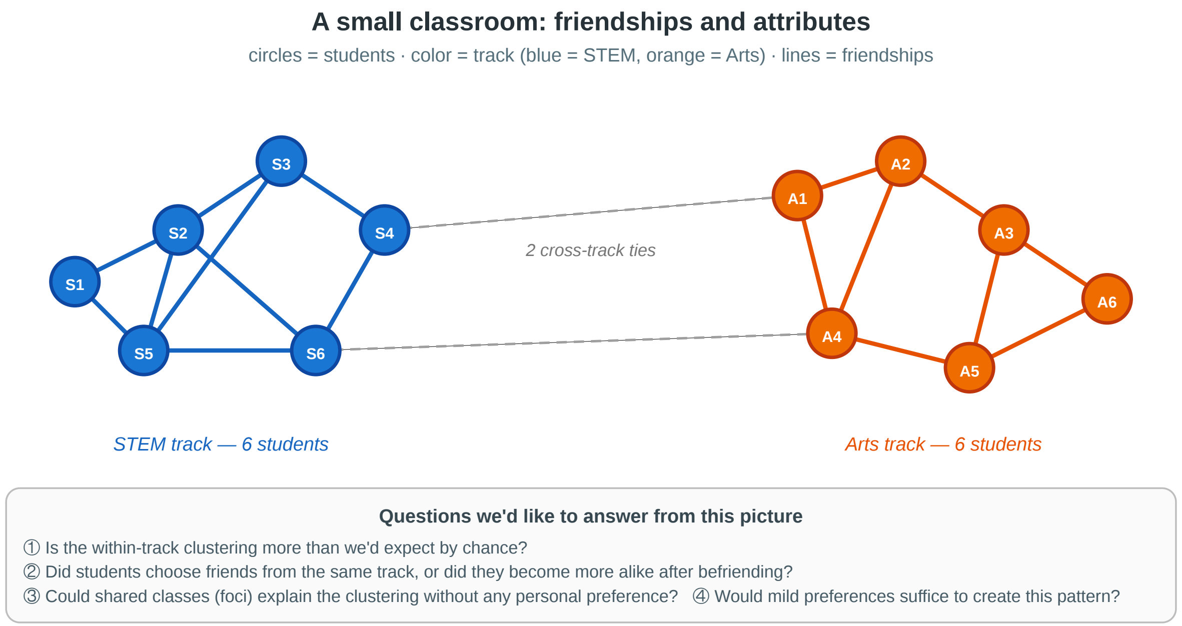
- 17 within-track edges of 19 total: $H_{\text{obs}} = 17/19 \approx 0.89$
- $r = (0.89 - 0.50)/(1 - 0.50) = 0.78$

$r = 0.78$ is strong — far above random mixing.

But it does not tell us *why*. Is it selection, socialization, or shared context?



Quick Check

In a school of 200 students, 60% are in track A and 40% in track B. Of all friendships, 78% are within-track.

1. Compute the random-mixing baseline H_{base}.
2. Compute the homophily index r.
3. Is r strong evidence of homophily? What could make it misleading?

Speaker notes

Give them 4 minutes before revealing the fragments. (1) $H_{\text{base}} = 0.6^2 + 0.4^2 = 0.52$. (2) $r = (0.78 - 0.52)/(1 - 0.52) = 0.26/0.48 \approx 0.54$. (3) $r = 0.54$ is moderate-to-strong, but it can be misleading if the tracks share different physical spaces (focal closure), if students were assigned to tracks based on pre-existing friend groups (reverse causation), or if family background causes both track choice and friendship patterns (confounding). Reveal the three answer fragments one at a time as students suggest theirs.

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3. $r = 0.54$ is moderate-to-strong evidence of homophily. It could be misleading if:

- the two tracks occupy different buildings (shared context, not personal preference);
- students were *assigned* to tracks based on pre-existing friend groups (reverse causation);
- a third factor (e.g. family background) causes both track choice and friendship patterns (confounding).

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Echo Chambers — The Hypothesis

Source: Cinelli et al. 2021

Claim	Online platforms expose users mostly to like-minded peers and sources.
Network signal	High homophily in the *interaction* network (likes, retweets, replies, comments) when nodes are coloured by political leaning.
Diffusion signal	Information circulates *within* groups rather than crossing them.

Why it's a homophily story. An echo chamber is not just "people who disagree online" — it is a region of the interaction graph where within-group ties dominate so heavily that the same content recirculates without leaving the cluster.

The empirical test (Cinelli et al. 2021, PNAS).

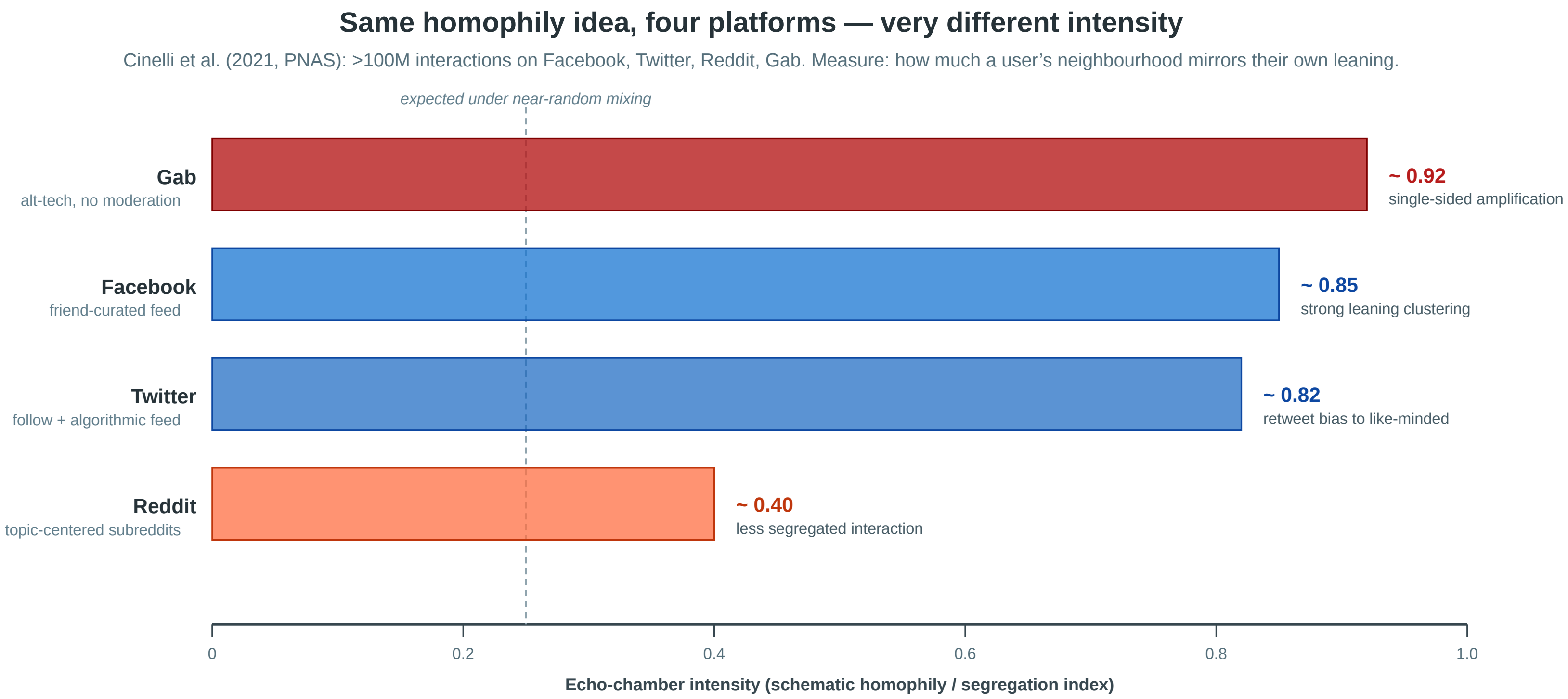
- Four platforms: **Facebook, Twitter, Reddit, Gab**.
- Over **100 million** interactions, hundreds of news outlets coded by political leaning.
- For each user i, infer a leaning score x_i as the **average leaning of the scored posts, links, pages, or sources** they produce or endorse; then compute the average leaning of their interaction neighborhood: $x_i^N = \frac{1}{k_i^{out}} \sum_j A_{ij} x_j$.
- If $x_i \approx x_i^N$, the user mostly interacts with like-minded peers.

So x_i is a behavioral estimate: it summarizes the political leaning of the content a user chooses to post, like, share, or link to. Different platform architectures (algorithmic feed, friend graph, subreddit topic, alt-tech) let us see whether echo chambers are universal or **architecture-dependent**.

Speaker notes

This opening slide gives the *network* framing for the echo chamber concept: it is a regional, homophily-based pattern in the interaction graph. The Cinelli et al. study is the largest cross-platform test of that claim. They operationalize echo-chamber intensity as how closely a user's interaction neighbourhood mirrors the user's own leaning. The four-platform design is what makes the study informative beyond Facebook — it lets the analyst distinguish "echo chambers exist online" from "echo chambers depend on how the platform is built."

Echo Chambers — Same Idea, Four Platforms



Schematic. Magnitudes are illustrative; the paper reports leaning-gap distributions, not a single index. The qualitative ranking (Gab > Facebook ≈ Twitter ≫ Reddit) matches the paper.

Architecture matters. Facebook and Twitter (friend / follow graphs amplified by ranking) show strong homophilic clustering by political leaning. **Gab** — small, unmoderated, single-sided audience — is even more extreme. **Reddit** is much less segregated: topic-centered subreddits force users to interact across leanings within the same subforum. The same homophily concept produces dramatically different intensities across designs.

Speaker notes

The qualitative ranking (Gab > Facebook ≈ Twitter ≫ Reddit) is the central empirical result. Use the figure to make two pedagogical points: (i) "echo chamber" is not a binary — it is a measurable intensity along the homophily continuum from $r = 0$ to $r = 1$; (ii) the intensity is a *property of the platform's architecture*, not a fixed feature of human behaviour. Friend-graph platforms with ranking-based feeds make echo chambers stronger; topic-graph platforms make them weaker. This connects directly to the affiliation-network module that follows: subreddits are explicit foci, friend feeds are not.

Takeaway

Homophily is not just a visual pattern — it is a statistical claim that tie formation is biased toward similarity *relative to a random-mixing baseline*. The homophily index r quantifies the excess, but it cannot identify the mechanism.

Selection and Socialization

> **Guiding Question:** How can we distinguish whether similarity caused the tie or the tie caused the similarity?

Speaker notes

This is arguably the most important conceptual module in the lecture. The three mechanisms — selection, socialization, and contextual correlation — produce the same cross-sectional pattern. Distinguishing them requires temporal data and careful causal reasoning. Without longitudinal observations, the $r = 0.78$ we just computed is compatible with any mixture of the three.

Selection and Socialization

> **Guiding Question:** How can we distinguish whether similarity caused the tie or the tie caused the similarity?

$$\text{homophily} = \text{selection} + \text{socialization} + \text{context}$$

Three Mechanisms, One Pattern

Selection. People choose to form ties with others who are *already* similar to them. Similarity precedes the tie.

Socialization (social influence). People become *more similar* to their contacts after the tie forms. The tie precedes similarity.

Contextual correlation. A shared environment (school, workplace, neighborhood) independently causes both the attribute and the tie. Neither causes the other directly.

Speaker notes

All three mechanisms are consistent with a high homophily index. A cross-sectional snapshot — friendships and attributes measured at one point in time — cannot separate them. This is the causal wall that L05 is built around.

Same Snapshot, Three Stories

Suppose we observe: **STEM students mostly befriend STEM students.**

Mechanism	Concrete story	What data would help?
Selection	Students first choose STEM, then choose friends from the same track.	Friendships observed *after* track choice
Socialization	Friends influence each other to choose the same elective or track.	Track changes observed *after* friendships
Contextual correlation	STEM students share labs, schedules, and buildings, so they meet more often.	Timetables, room assignments, shared courses

> The graph snapshot is identical in all three stories. The difference is the **time order** and the **unobserved context**.

Speaker notes

Use this slide to make the causal distinction tangible before showing the causal diagrams. The point is not that the mechanisms are mutually exclusive; in real schools all three can operate. The task is to ask which data would let us estimate their relative contribution.

Causal Diagrams

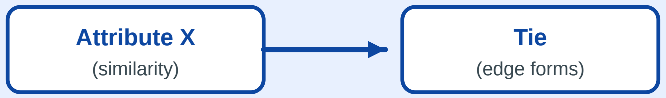
Three causal mechanisms — same observed pattern

All three produce the cross-sectional fact “connected nodes are similar.” Distinguishing them needs temporal data or knowledge of hidden context.

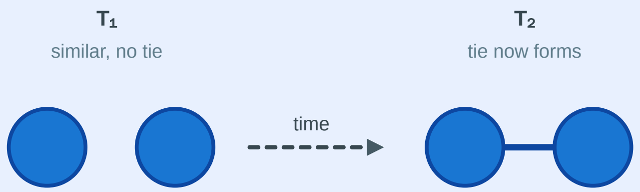
A. Selection

similarity → tie

Causal diagram



What happens over time



Classroom example

Two students are already in STEM. They meet and become friends *because* they share that interest.

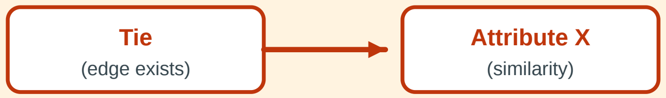
Empirical signature:

new ties form preferentially between already-similar people.

B. Socialization (influence)

tie → similarity

Causal diagram



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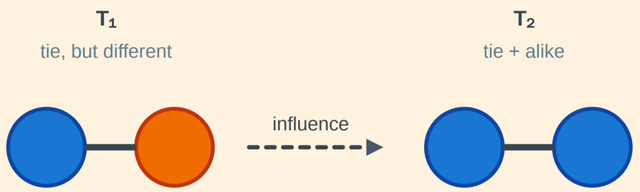


Classroom example

A STEM and an Arts student become friends. Over months, the Arts student shifts interests toward STEM.

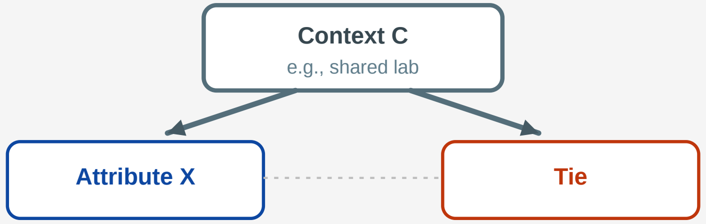
Empirical signature:

attributes change after ties form, in the direction of friend's attribute.

C. Contextual correlation

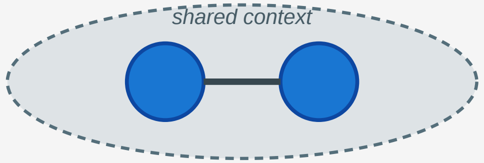
C → similarity AND C → tie

Causal diagram



no direct arrow between X and Tie

What we see in the data



Classroom example

Two students share a lab. The lab independently makes them similar **and** brings them together.

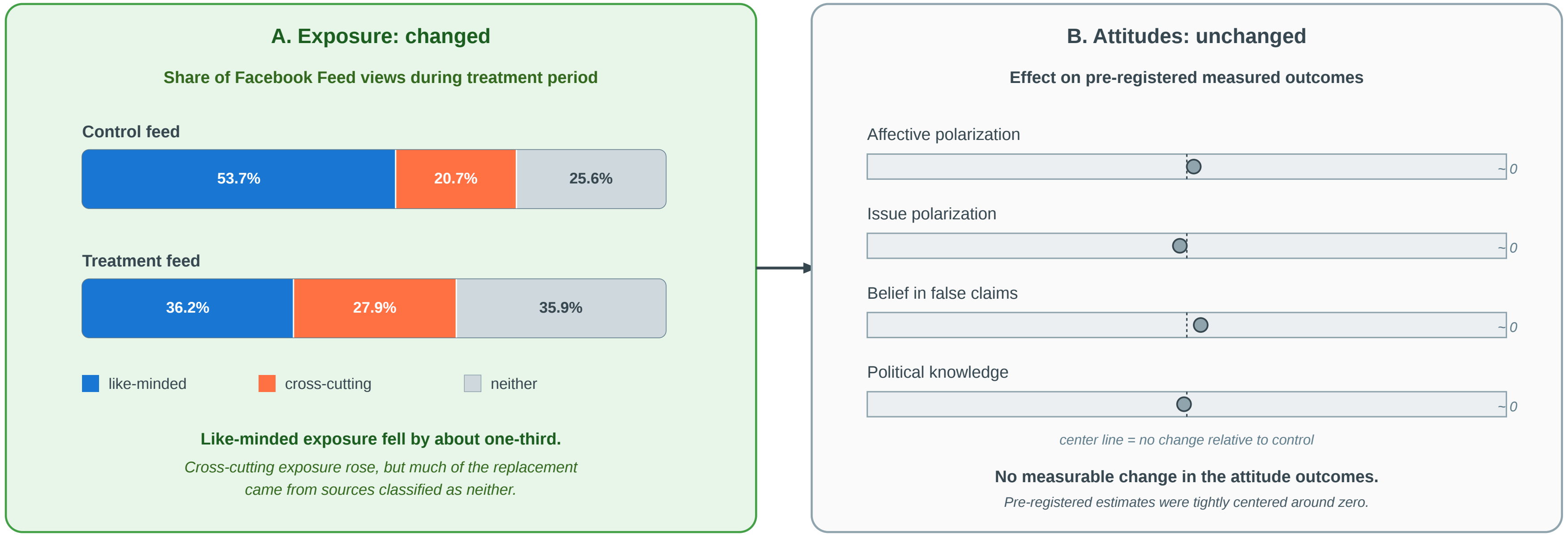
Empirical signature:

association vanishes when conditioning on C.

Speaker notes

Walk through the three causal diagrams: selection (similarity → tie), socialization (tie → similarity), context (third factor → both). Without temporal data, all three explain the same cross-sectional correlation. The point is not that one is always right; in real networks, all three usually operate simultaneously. The analyst's job is to measure their relative contributions.

Empirical Test: Does Exposure Cause Polarization?



Source: *Nyhan et al. 2023*

Selection question: do already-polarized users choose like-minded friends, pages, and platforms? **Socialization question:** does repeated exposure to like-minded content make users more polarized?

Nyhan et al. (2023, Nature) isolate the second pathway. In a pre-registered Facebook field experiment during the 2020 U.S. election, **23,377 consenting users** were randomly assigned to a feed intervention that reduced politically like-minded source exposure by about one-third. The treatment increased cross-cutting exposure, but had **no measurable effect** on eight pre-registered attitude outcomes, including affective polarization, ideological extremity, candidate evaluations, and belief in false claims.

Interpretation. The experiment weakens one short-run **socialization** story: changing feed exposure for three months did not measurably change attitudes. It

Speaker notes

This belongs in the selection/socialization module because it cleanly separates two mechanisms. Selection means politically polarized users may actively choose like-minded friends, pages, groups, and platforms. Socialization means exposure itself changes attitudes after the tie or feed relation exists. Nyhan et al. randomize exposure, so the study directly tests the socialization path from feed content to attitudes. It does not randomize the user's underlying network or platform choice, so it cannot rule out selection-based echo-chamber formation. The null result is therefore informative but bounded: cross-cutting exposure increased, yet the measured attitude outcomes did not move detectably.

Empirical Test: Christakis & Fowler — Obesity in Social Networks

Christakis & Fowler (2007), *The Spread of Obesity in a Large Social Network Over 32 Years*, New England Journal of Medicine 357, 370–379.

They analyzed the **Framingham Heart Study** social network — 12 067 people tracked over **32 years** with repeated measurements of both friendship ties and body mass index. This made it possible to observe the *temporal sequence* of tie formation and weight change.

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Speaker notes

The Framingham study is the canonical longitudinal analysis of selection vs. socialization. Its strength is the multi-decade time series: because both the network and the attribute (BMI) were measured repeatedly, the authors could observe the temporal order — who became obese *before* whom — and test whether the pattern fit selection, socialization, or confounding. The 57% figure is a relative risk increase among nominated friends compared to non-friends, controlling for geographic proximity.

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> **Christakis & Fowler (2007),** *The Spread of Obesity in a Large Social Network Over 32 Years*, New England Journal of Medicine 357, 370–379.
>
> They analyzed the **Framingham Heart Study** social network — 12 067 people tracked over **32 years** with repeated measurements of both friendship ties and body mass index. This made it possible to observe the *temporal sequence* of tie formation and weight change.

Headline result. A person's probability of becoming obese increased by **57%** if a friend became obese in the same interval. The effect was stronger for mutual friends and showed **directionality** — it operated in the direction of the friendship nomination, suggesting socialization (influence) rather than pure selection.

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What the Study Could Not See

1. Confounders. Cohen-Cole & Fletcher (2008) and Lyons (2011) showed that the same statistical method would "detect" social contagion of **height** and **acne** — traits that clearly do not spread through social influence. The method may confuse *shared environment* with *social influence* when the confounders evolve over time.

2. Dynamic selection. The study controlled for whether two friends were already similar in weight. But people may also **choose new friends**, or keep reporting old friends, because their lives and weight trajectories are becoming similar. Then "my friend became obese before me" may partly reflect changing friendship selection, not only social influence.

3. Missing network. The Framingham data records nominated friends, not the full social network. Influence may flow through unobserved ties (coworkers, family, neighbors) that correlate with both friendship nominations and weight change.

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Speaker notes

This slide is the L05 analogue of L03's "what Kossinets-Watts could not see" and L04's "what Zachary could not see". The pattern is deliberate: each lecture celebrates an empirical result and then identifies its blind spots. The Framingham critique is not a rejection of the study — it is a reminder that even 32-year longitudinal data with repeated measures cannot eliminate all causal alternatives. The lesson is humility, not nihilism.

Quick Check

You observe that smokers tend to befriend other smokers in a college dorm.

1. What kind of data would you need beyond a single snapshot to distinguish selection from socialization?
2. Sketch one observation pattern that would favor *selection* and one that would favor *socialization*.
3. Even with longitudinal data, what kind of confound would remain hard to rule out?

Speaker notes

Give them 4 minutes before revealing the fragments. (1) Longitudinal data: friendships and smoking status at multiple time points. (2) Selection: new ties form between people who already both smoke. Socialization: non-smokers start smoking after befriending smokers. (3) Shared environment: dorm floor placement may independently cause both friendship formation and smoking uptake (e.g. shared common rooms where smoking happens).

Quick Check

You observe that smokers tend to befriend other smokers in a college dorm.

1. What kind of data would you need beyond a single snapshot to distinguish selection from socialization?
2. Sketch one observation pattern that would favor *selection* and one that would favor *socialization*.
3. Even with longitudinal data, what kind of confound would remain hard to rule out?

1. Longitudinal observations — friendships and smoking status at multiple time points, with newly formed ties and newly adopted behavior visible.

2.

- *Selection:* new friendships disproportionately form between people who **already** share smoking status.
- *Socialization:* people who befriend smokers become more likely to **start** smoking in the next observation period.

3. Shared *environment* — e.g. dorm floor placement causing both friendship and smoking exposure simultaneously. Even with temporal data, confounders that evolve over time alongside both the network and the attribute are hard to rule out without randomization.

Takeaway

> Cross-sectional data cannot distinguish selection from socialization — the same homophily index r is consistent with both. Longitudinal network data is far more informative, but even multi-decade panel studies cannot fully eliminate confounding by evolving shared environments.

Affiliation Networks

> **Guiding Question:** How do we represent the social settings — classes, clubs, workplaces — in which ties form?

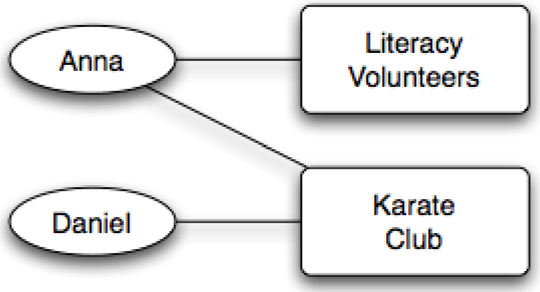
Speaker notes

Affiliation networks make the "shared context" mechanism from Module 2 explicit and formal. Instead of treating context as an unobserved confounder, we model it directly: people are connected to foci, and ties between people arise through their shared foci. This is the modeling answer to Question 3 from the opening slide.

Bipartite Affiliation Model

An **affiliation network** is a bipartite graph $G = (P \cup F, E)$:

- P: actor nodes (persons)
- F: focus nodes (groups, locations, classes, organizations)
- E: edges only between P and F — "person p participates in focus f"



Source: Easley & Kleinberg 2010

The focus nodes are the **social context**: they encode where people meet, what they jointly attend, and which settings create repeated exposure. This lets us separate "people are similar" from "people share the same places."

Shared foci do not prove a tie. They create opportunities for contact, information, and influence.

Speaker notes

This connects back to Lecture 02's bipartite graphs. The bipartite structure is a modeling decision: we choose to represent contexts as a separate node type rather than as edge attributes. The payoff is that closure processes become explicit and testable. The classroom scenario can be modeled this way: the two tracks are foci, and within-track friendships arise partly through shared-focus exposure.

From Affiliations to Co-Occurrence Graphs

The bipartite graph is the source of truth. To use standard network tools, we often project it into a one-mode **co-occurrence graph**.

Let B be the person-by-focus membership matrix:

$$B_{pf} = 1 \quad \text{if person } p \text{ participates in focus } f.$$

Then:

- **Person co-occurrence:** $BB^\top$
- **Focus co-occurrence:** $B^\top B$

An edge weight counts **how many contexts are shared**.

Projection	Nodes	Edge weight means	Then analyze with
$BB^\top$	People	shared classes, clubs, boards, workplaces	centrality, communities, clustering
$B^\top B$	Foci	overlapping participants	similarity, clusters, diffusion channels

> Co-occurrence is **context**, not a confirmed relationship. It says two units had an opportunity to interact.

Speaker notes

This slide is the missing bridge between the affiliation model and ordinary graph analysis. The bipartite graph records membership directly; the projected co-occurrence graph turns shared contexts into weighted edges so that familiar tools such as centrality, clustering, and community detection can be used. Stress the warning: projection is useful, but it changes the object of analysis. A person-person edge in $BB^\top$ means shared context, not necessarily friendship.

Reading the Board Network

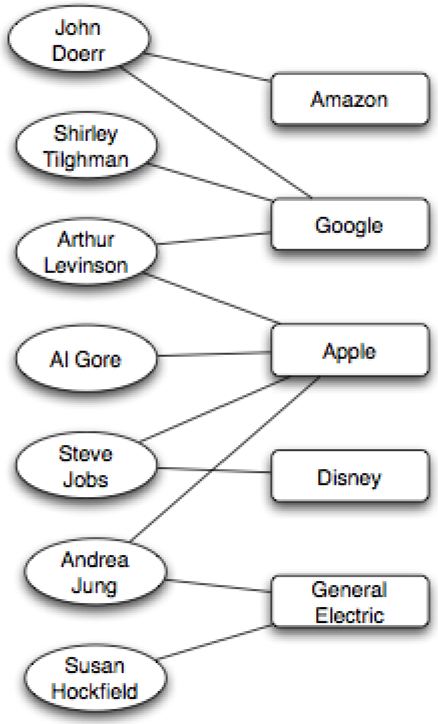
Corporate boards are an affiliation network: board members connect to company boards. The co-occurrence projection decides what counts as a node and what the shared context means.

Co-occurrence graph	Edge means	Use it for
Board-member graph	Two board members share ≥ 1 company board	Peer exposure, governance norms
Company graph	Two firms share ≥ 1 board member	Strategic interlocks, diffusion across firms

Reading rules.

1. A projected edge means **shared context**, not friendship or influence.
2. Projection creates dense cliques: a board with s members adds $\binom{s}{2}$ member-member edges.
3. Keep weights: sharing five boards is stronger evidence than sharing one.

Example — Corporate Board Interlocks

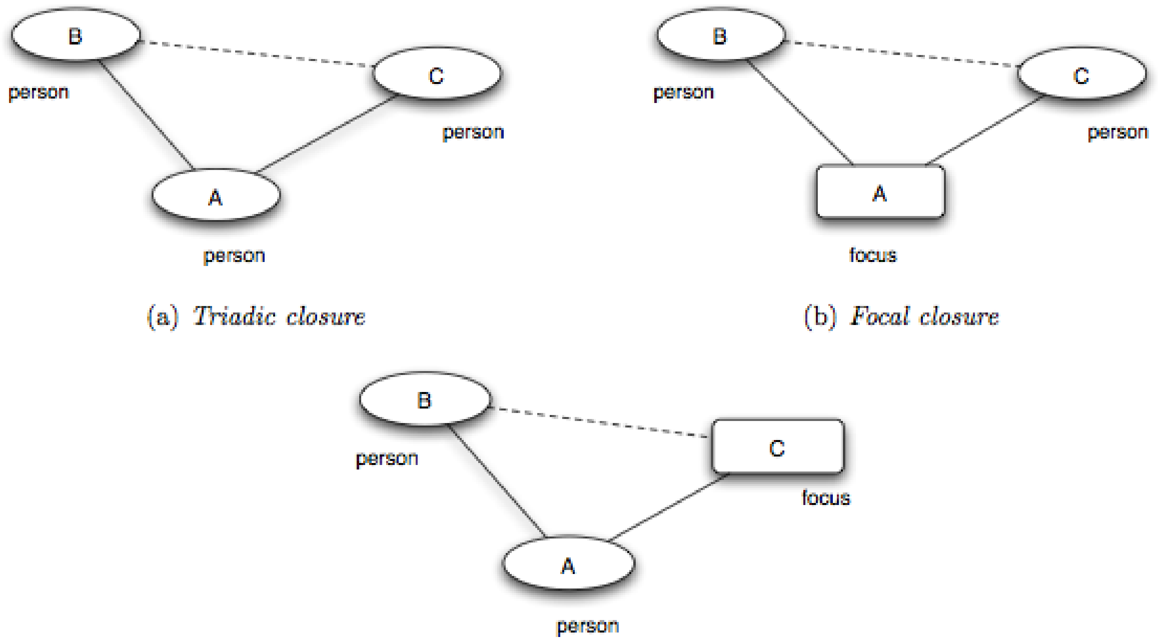


Source: Easley & Kleinberg 2010

Speaker notes

This slide applies the previous co-occurrence idea to corporate boards. The choice of projection is not neutral: it foregrounds either board-member exposure or firm-level interlocks. Emphasize the information-loss point: after projection, an edge no longer tells you which board mediated the tie unless the weight or original bipartite data is kept. Practical consequences: projections create dense cliques, and edge weights are usually informative. The exposure-not-friendship caveat is the same point as in diffusion and contagion: co-membership is a channel, not yet a cause.

Three Closure Processes



Source: Easley & Kleinberg 2010

Triadic closure. Friend-of-friend becomes friend — a person–person–person path closes into a triangle. *(Same mechanism as Lecture 03.)*
Focal closure. Two persons sharing a focus become friends — a person–focus–person path closes into a person–person tie. *(New — the focus mediates the introduction.)*
Membership closure. A friend introduces you to a focus you did not belong to — a person–person–focus path closes into a person–focus tie. *(New — the friend mediates membership.)*

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Speaker notes

All three are mechanisms that turn structure into *more* structure. Triadic closure was L03's core concept; focal and membership closure are L05's additions. In the classroom figure, within-track friendships could arise through focal closure (shared labs and lectures bring STEM students together), and if an Arts student (A1) befriends a STEM student (S4) and later joins a STEM elective, that is membership closure. The distinction matters because focal and membership closure produce homophily-like patterns *without* any personal preference for similarity.

Closure Processes — Classroom Examples

Closure type	Open path	What closes?	Classroom example
Triadic	student → friend → student	A new friendship	"My lab partner introduces me to their friend."
Focal	student → elective → student	A new friendship	"We sit in the same statistics elective and start talking."
Membership	student → friend → club	A new membership	"My friend invites me to join the robotics club."

> Only triadic closure needs a shared friend. Focal closure can create ties among strangers through repeated exposure in the same setting.

Speaker notes

This table gives students a concrete classification handle before the quick check. Emphasize that focal closure is often enough to create observed homophily: people in the same track share many foci, so within-track ties become more likely even without an explicit preference for same-track friends.

Quick Check

For each scenario, identify whether triadic, focal, or membership closure is at work.

1. Two students take the same elective and become friends.
2. A new graduate student joins the same lab as her advisor's other student, who introduces her.
3. A friend invites you to their book club, and you join.

Speaker notes

Give them 3 minutes before revealing the fragments. (1) Focal closure — the elective is the shared focus. (2) Triadic closure — the new friendship forms via a shared friend (the advisor). (3) Membership closure — your friend introduces you to a focus you did not previously belong to.

Quick Check

For each scenario, identify whether triadic, focal, or membership closure is at work.

1. Two students take the same elective and become friends.
2. A new graduate student joins the same lab as her advisor's other student, who introduces her.
3. A friend invites you to their book club, and you join.

1. Focal closure — the elective is the shared focus that brings them into contact.

2. Triadic closure — the advisor is the shared friend that closes the triangle.

3. Membership closure — your friend brings you into a focus you did not previously belong to.

Takeaway

> Affiliation networks make social context explicit. Focal closure and membership closure are distinct tie-formation mechanisms that can produce homophily-like patterns without any personal preference for similarity — making them a critical alternative explanation in any homophily analysis.

Online Link Formation

Guiding Question: What can large-scale online data teach us about how ties form over time?

Speaker notes

Online platforms make link formation a measurable process at scale. Every new edge has a timestamp, and every preceding state is observable. This is what makes large-scale empirical tests of closure possible — and it is also where we finally get the longitudinal data that Modules 1–2 argued we need.

The Setting — Kossinets & Watts (2006)

Data. A large U.S. research university's internal email logs, ca. 1 year, **~43,000 students/faculty/staff**, **~14 million** messages — plus full course-enrollment records and organizational-affiliation rosters.

Why it matters. The timestamps make the analysis forward-looking: we can observe what existed at time t, then ask which new email ties form at $t + 1$.

Used here for two experiments.

- **Triadic closure:** shared email contacts.
- **Focal closure:** shared classes / institutional foci.

The membership-closure plots that follow are separate Easley-Kleinberg examples from LiveJournal and Wikipedia, not the same Kossinets-Watts university dataset.

Speaker notes

Flag the setting before showing the triadic and focal closure curves. The Kossinets-Watts university data gives email timestamps plus course and organizational affiliation data, so it can support both shared-friend and shared-focus analyses. Do not imply that the later membership-closure figures use this same dataset; those are separate Easley-Kleinberg examples from LiveJournal and Wikipedia.

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> The timestamped data separates *"what existed at t"* from *"what formed by $t+1$"* — the missing ingredient in homophily snapshots.

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Revisit Triadic Closure first: Independent-Opportunity Baseline

Before reading the empirical curve, we should think of a baseline. What could that be?

Speaker notes

Introduce this before the Kossinets-Watts triadic closure plot because the red dotted lines in the figure are independent-opportunity baselines. The formula is a complement rule: closure happens if at least one of the k mutual friends creates an opportunity. If one mutual friend fails to create an opportunity with probability $1-q$, and the k opportunities are treated as independent, then all k fail with probability $(1-q)^k$. Subtracting from 1 gives the probability that at least one opportunity succeeds. The pedagogical table uses $q=0.20$ because the concavity is easy to see. In the empirical plot, however, q is roughly 4×10^{-4}, so $1-(1-q)^k \approx kq$ for the shown k values; the curvature is mathematically present but visually negligible. Easley and Kleinberg plot this curve plus a one-step-shifted version, roughly $1-(1-q)^{k-1}$ for $k \geq 1$, because the first common friend has a small absolute effect in the data. This is intentionally simple. It provides a reference shape against which the empirical black curve can be read.

Revisit Triadic Closure first: Independent-Opportunity Baseline

Before reading the empirical curve, we should think of a baseline. What could that be?

Assume each mutual friend independently creates an opportunity for a new tie with probability q.

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Assume each mutual friend independently creates an opportunity for a new tie with probability q.

$$P(\text{new tie} \mid k \text{ shared friends}) = 1 - (1-q)^k$$

Shared friends k	1	2	3	5
P if $q = 0.20$	0.20	0.36	0.49	0.67

Why this form?

- q: probability, one mutual friend creates an opportunity.
- $1 - q$: that mutual friend does **not** create one.
- $(1-q)^k$: none of the k mutual friends creates one.
- $1 - (1-q)^k$: at least one opportunity appears.

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- $(1 - q)^k$: none of the k mutual friends creates one.
- $1 - (1 - q)^k$: at least one opportunity appears.

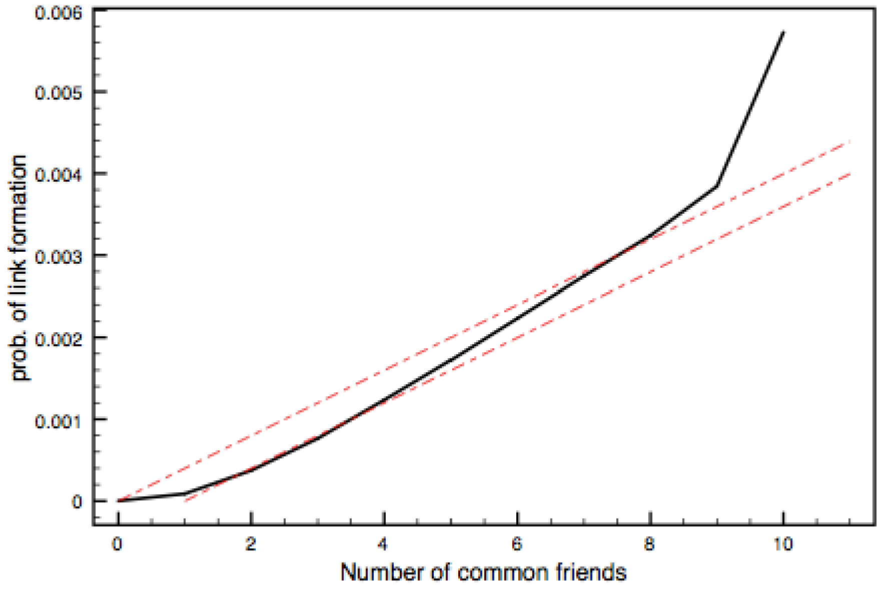
What does this mean? The model gives us a **comparison curve**, not a claim that friends really act independently. The table uses a large q only to make the curvature visible. In the Easley-Kleinberg figure, q is tiny, so the curve looks almost linear on the plotted range.

In the empirical figure, the second dotted curve is the same baseline shifted one step right: it asks what the curve would look like if the first mutual friend had a smaller effect.

Triadic Closure at Scale

Black curve: observed probability $T(k)$. Red dotted curves: independent-opportunity baselines from the previous slide. (Upper/Lower: $1-(1-q)^k; 1-(1-q)^{k-1}; q \approx 4 \times 10^{-4}$)

Because q is very small here, $1-(1-q)^k \approx kq$ over the plotted range, so the dotted baseline appears nearly linear even though the exact function is concave.



Source: Easley & Kleinberg 2010

What was measured. At each time step, take every pair (u, v) that has **not yet emailed**. Count k: how many mutual email contacts they already share.

Then ask whether (u, v) exchanges email in the next interval.

The output is a curve: $P(\text{new tie})$ vs. k.

k shared friends	$P(\text{new tie})$
0	very low (baseline)
1	sharply higher
2	higher still
5--8	roughly follows the independent-friend baseline
9+	rises steeply again, but with fewer observations

1. **Mechanism.** Mutual friends create *opportunities*: introductions, shared interests, social pressure to know each other.
2. **Shape.** The curve is not just "more friends, linearly more probability." It is close to the independent baseline in the middle, but bends upward at high k; those high-k points should be read cautiously because there are fewer pairs with many mutual friends.

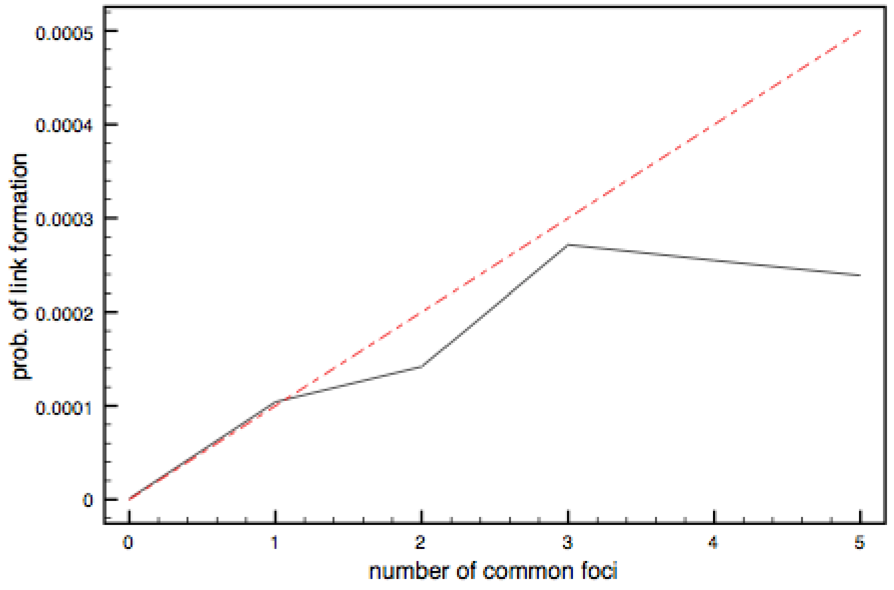
Do not read the red-dotted lines as confidence intervals. They are toy-model baselines, not two fitted values of q. The value of q only puts the comparison curve on the observed scale; it is not a causal estimate of one mutual friend's effect. Also be cautious at high k: pairs with many mutual friends are rare, so the curve can be noisier there.

Speaker notes

The empirical curve is the central evidence for triadic closure at scale. Walk students through three landmarks: (i) the baseline rate at $k = 0$ — essentially how often two people with no mutual contacts start emailing — is very small; (ii) the probability rises strongly once mutual friends exist; (iii) the red dotted curves are independent-opportunity baselines, not confidence intervals. Easley and Kleinberg do not use two different q settings here; they plot the simple baseline and a one-friend shifted version. From the y-axis scale, q is roughly 4×10^{-4} per shared friend per day. The black curve is close to this baseline through the middle range but bends upward for very high k. Avoid describing this slide as simple saturation; the high-k rise is visible, although it is also based on rarer pairs.

Focal Closure at Scale

Black curve: observed probability. Red dotted curve: independent-focus baseline, $T_{\text{base}}(k) = 1 - (1-p)^k$, with p calibrated from the effect of one shared class.



Source: Easley & Kleinberg 2010

Setup. Two people who have **never emailed each other** but co-enroll in k classes during the same semester. What is the probability they exchange email within the term?

Result. One shared class is already informative. But after that, the observed black curve stays **below** the independent-focus baseline: additional shared classes add less than the toy model predicts.

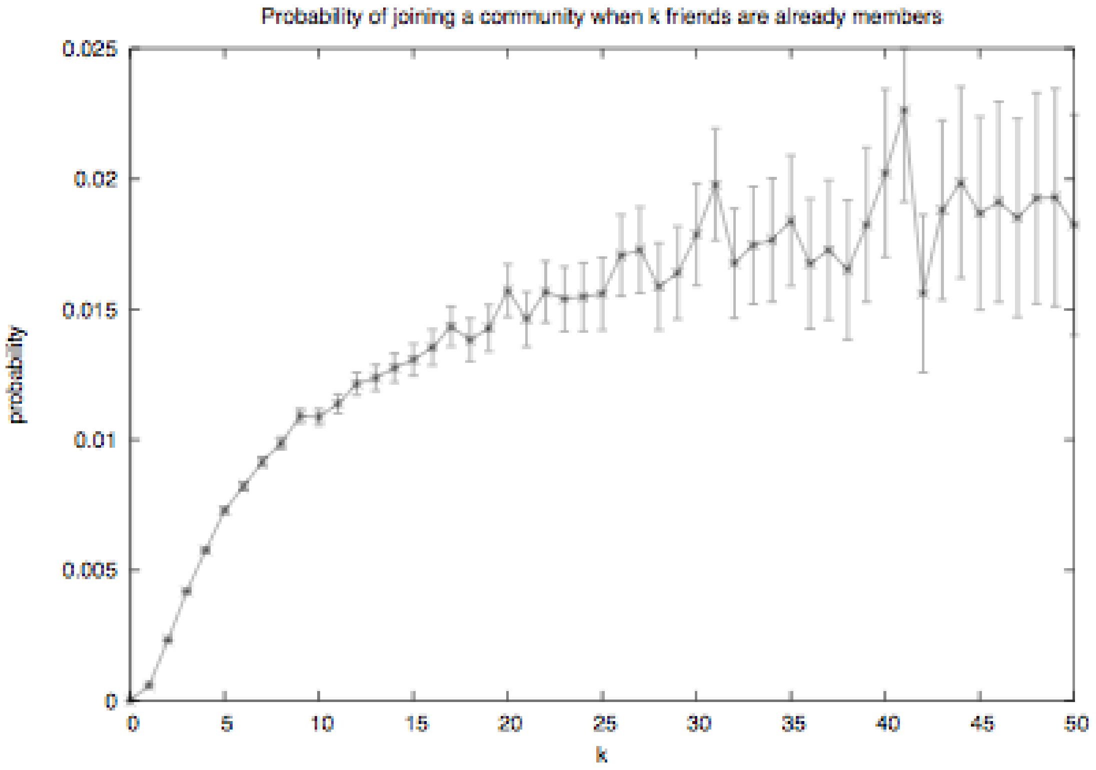
- Focal closure runs **without any shared friend**. Strangers in the same context form ties through repeated exposure alone.
- This is the mechanism that turns "people share classes" into "people share friendships" — and it produces homophily-like patterns even when there is **no preference for similarity at all**.
- It is L05's main answer to *Question 3* from the opening slide ("could shared context explain the clustering?"). Often, yes.

Interpret the dotted line as a baseline, not a fitted causal law. The gap below it means shared classes are partly redundant: taking five classes together is not five independent chances to become connected.

Speaker notes

Focal closure is L05's distinctive concept (triadic was L03's). The dotted curve is the same independent-opportunity baseline used for triadic closure, now with shared classes as the opportunities: $T_{\text{base}}(k) = 1 - (1-p)^k$. Since $T_{\text{base}}(1) = p$, p can be read from the one-shared-class effect. The important empirical difference from triadic closure is that the focal-closure curve turns downward and levels off below the independent baseline. Subsequent shared classes after the first have diminishing returns. The teaching link to homophily is direct: if STEM students share many classes, focal closure alone can generate a homophily-rich within-STEM friendship network without any selection effect.

Membership Closure at Scale



Source: Easley & Kleinberg 2010

Setup. A person is **not yet a member** of a focus (course, club, group). How does the probability of joining depend on the number of existing members who are already their friends?

Result. Same saturating shape. One friend already in the focus is a strong predictor of joining; additional friends compound — but with diminishing returns.

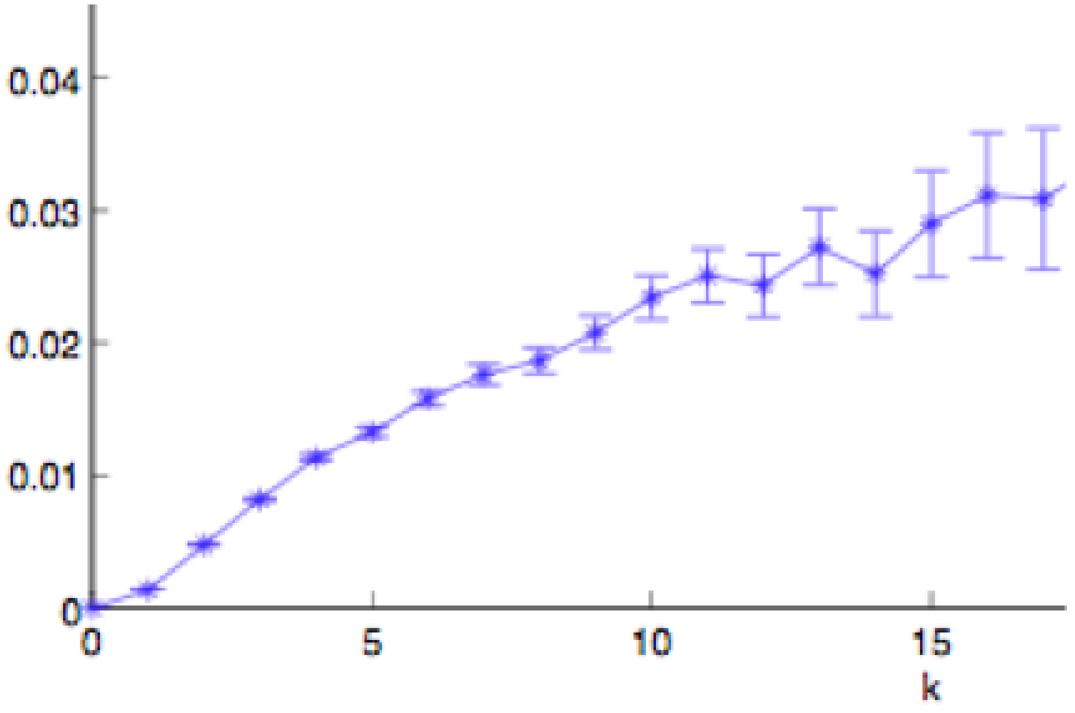
The flipped direction.

- Triadic and focal closure close *person–person* ties.
- Membership closure closes a *person–focus* tie — your friends pull you **into** the contexts they already inhabit.
- This is how **social networks shape affiliations**, not just the other way around. Friendships and foci co-evolve through both directions of closure.

Speaker notes

Membership closure flips the bipartite arrow: instead of foci producing person–person ties (focal closure), person–person ties produce focus-memberships. This LiveJournal figure is a separate Easley-Kleinberg example, not the Kossinets-Watts university data. In the classroom example, if an Arts student A1 has multiple STEM friends, A1 is more likely to take a STEM elective. This is how friendships drive affiliations, completing the co-evolution loop between the network and the foci.

Wikipedia Editing as Membership Closure



Source: Easley & Kleinberg 2010

Dataset. Wikipedia edit histories plus editor communication. Editors are people; articles are foci; user-talk interactions define editor-editor social ties.

Setup. An editor is connected to an article after editing it.

For an editor who has **not yet edited** an article, count k: how many of their communication partners have already edited that article?

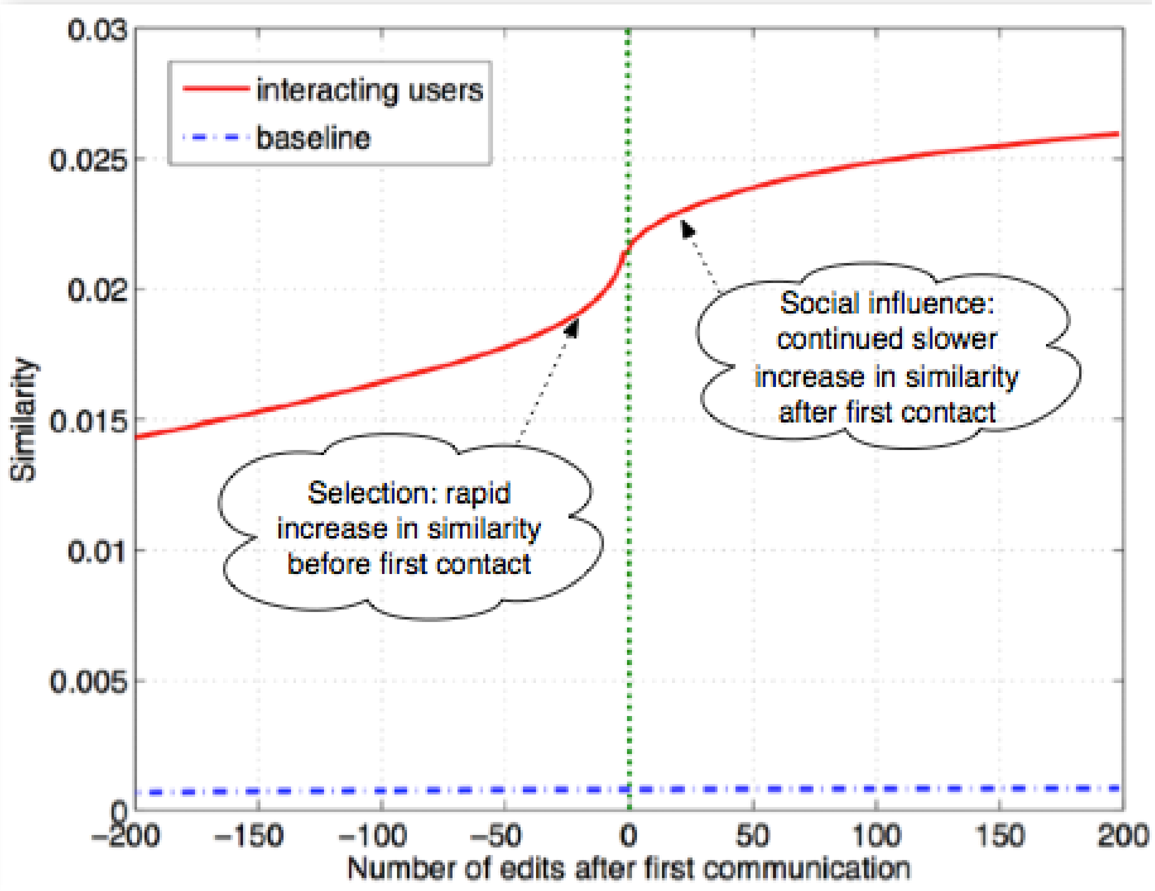
Result. The probability of editing the article rises with k. Friends already active in a focus make joining that focus more likely, with diminishing marginal returns.

> This is membership closure with a shared knowledge object as the focus: friends already active around an article make later editing of that article more likely.

Speaker notes

This is the Wikipedia experiment discussed by Easley and Kleinberg. The social network is editor-editor communication via user talk pages; the affiliation network connects editors to articles they have edited. The plot asks whether an editor is more likely to edit an article when more of their communication partners have already edited it. The answer is yes: the probability rises with the number of friends already associated with the article, while marginal gains diminish.

Wikipedia Editors: Selection and Socialization



Source: Easley & Kleinberg 2010

Dataset. Wikipedia edit histories plus first user-talk communication between pairs of editors (Crandall et al. 2008).

Design. Align editor pairs at time 0: the moment they first communicate. Then track similarity in edited articles before and after that first contact.

Reading.

- Before first contact, future communication partners already become more similar than baseline: **selection**.
- After first contact, similarity continues rising more slowly: possible **socialization**.
- This is a feedback loop: similarity makes interaction more likely, and interaction can further increase similarity.

The figure separates timing, not perfect causality. It strongly suggests both selection and socialization, but unobserved shared contexts can still contribute to the pattern.

Speaker notes

This is the second Wikipedia example from Crandall et al. (2008). It is different from the previous membership-closure slide: the previous slide asks whether friends pull an editor into an article; this slide asks how editor-editor similarity changes around the moment of first communication. The steep increase before first communication is selection: similar editors are more likely to encounter or contact each other. The slower increase after first communication is consistent with socialization: once editors interact, their future editing patterns become more similar. The important teaching point is the timing logic: measuring similarity around first contact helps separate "similar people meet" from "people become more similar after meeting," though it does not remove every possible confound.

Quick Check

A study finds that the probability of two coworkers becoming Facebook friends rises from 1% (no shared friends) to 20% (one shared friend) to 30% (two) to 33% (three or more).

1. Describe the shape of the curve.
2. Fit the independent-opportunity model: what value of q gives $P = 0.20$ at $k = 1$?
3. Does the model predict the observed $P = 0.30$ at $k = 2$? What does the mismatch tell you?

Speaker notes

Give them 4 minutes before revealing the fragments. (1) Steep initial rise, then flattening. (2) $q \approx 0.20$ at $k = 1$. (3) The model overpredicts at $k = 2$; the redundancy among shared friends explains the gap. Connect this back to L04's embeddedness idea: shared friends are often embedded in the same dense cluster, so each additional shared friend gives less *new* opportunity.

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1. Describe the shape of the curve.
2. Fit the independent-opportunity model: what value of q gives $P = 0.20$ at $k = 1$?
3. Does the model predict the observed $P = 0.30$ at $k = 2$? What does the mismatch tell you?

1. Steep initial rise from 1% to 20%, then much smaller gains (30%, 33%) — a **saturating** curve.

2. At $k = 1$: $1 - (1 - q) = q$, so $q \approx 0.20$.

3. The model predicts $1 - (1 - 0.20)^2 = 0.36$ but we observe **0.30** — a shortfall. Shared friends are *correlated* (typically embedded in the same group), so they provide less independent opportunity than the model assumes. Diminishing returns are *steeper* than independence predicts.

Takeaway

Online data turns link formation into a measurable, time-stamped process. Empirical closure curves show saturation — the first shared contact matters most, and additional contacts are redundant rather than independent.

From Local Preferences to Global Segregation

Guiding Question: Can weak local preferences generate strong global segregation?

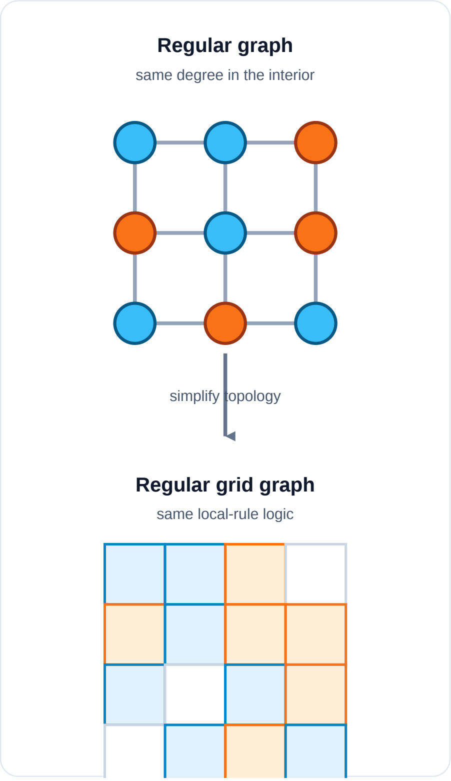
Speaker notes

This module answers Question 4 from the opening slide: "Would mild preferences suffice to create the sharp pattern we see?" Schelling's threshold model shows that the answer is yes — and the mechanism is iterated local adjustment, not central planning.

Segregation as a Graph

> **Segregation** means that groups are separated across neighborhoods or network regions: most local contacts are same-type, and cross-type contact is rare. It is a **global pattern** in the whole graph, not just one person's preference.

Schelling turned homophily into a dynamic model: **how can small local preferences for similar neighbors accumulate into large-scale segregation?** The model is useful because it separates a local rule from the global pattern it produces.



> **Model:** The graph version makes the mechanism explicit:
> - **nodes** are agents or households
> - **edges** are neighborhood/contact relations
> - a **grid** is a regular graph: each interior cell has the same local neighborhood size
> - **node type** is the attribute that matters locally
> - **updating** means moving on a grid or rewiring ties in a network

Speaker notes

Use this as the motivation for Schelling before introducing the threshold rule. The model belongs in a network science lecture because it turns "neighborhood" into an adjacency relation. A square grid is a regular graph topology: households are nodes, adjacent cells are edges, and most cells have the same number of neighbors. In online or organizational settings, there may be no physical grid; the same logic becomes rewiring, recommendation, or group switching. This connects the module back to the whole L05 arc: homophily is a measured local tendency, segregation is the macro-pattern, selection/socialization are causal mechanisms, and Schelling is a generative model showing how local rules can amplify into global structure.

Schelling's Threshold Model

Setup. Agents of two types are arranged on a grid (or network). Each agent i has a threshold $\tau \in [0, 1]$ — the minimum fraction of same-type neighbors it needs to be "satisfied".

Update rule.

1. Identify all *unsatisfied* agents (fewer than fraction τ of neighbors are same-type).
2. Each unsatisfied agent moves to a random vacant cell (or rewires a random tie in a network setting).
3. Repeat until no agent wants to move — or until a stable oscillation is reached.

Speaker notes

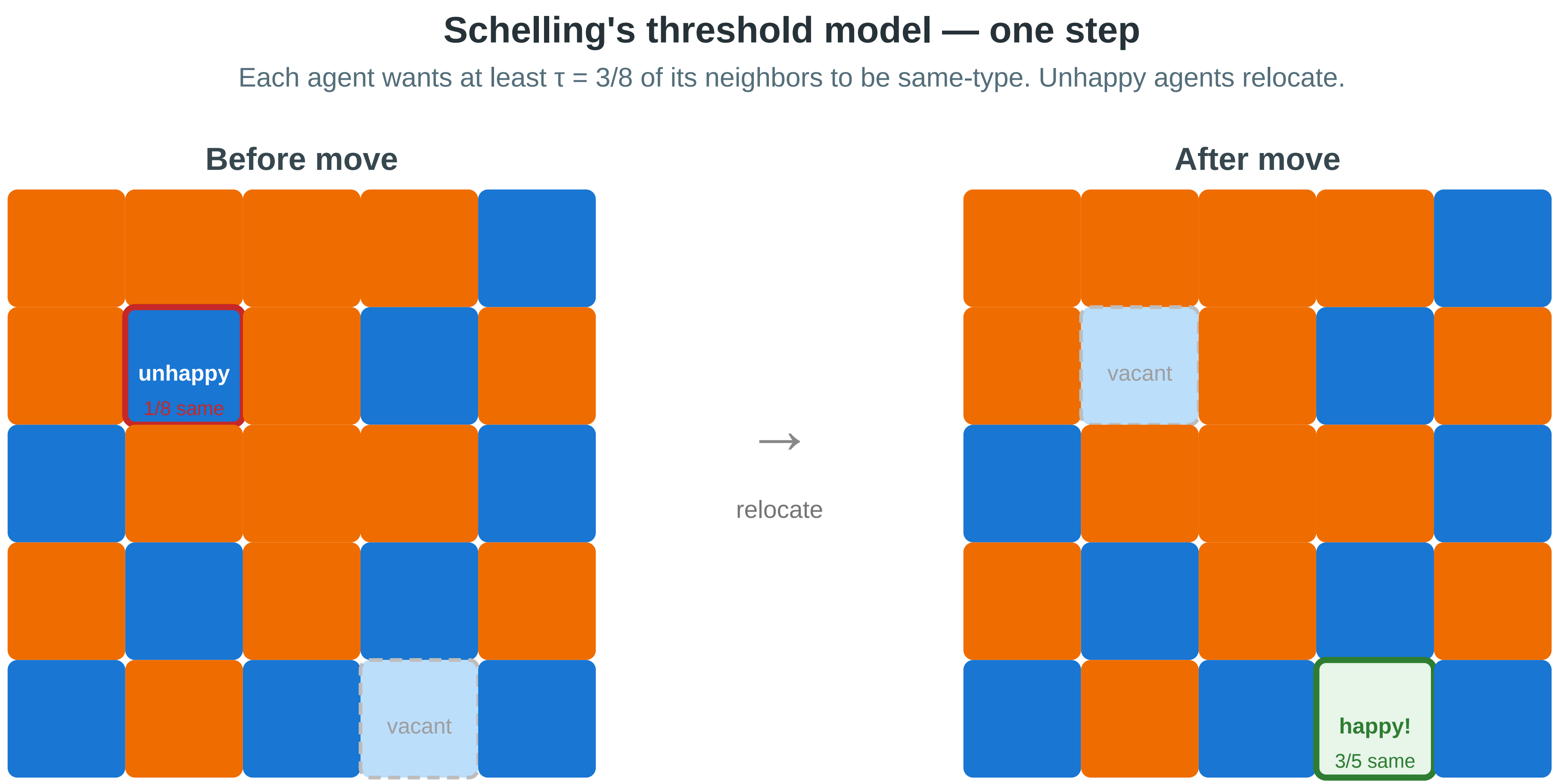
The formal model is simple enough to simulate in a few lines of code, yet its output is dramatically disproportionate to its input. The agent asks for 33% same-type neighbors; the iterated dynamics deliver 90%+ segregation. No agent *wanted* the global outcome; it emerged from the accumulation of individually mild adjustments.

Schelling's Threshold Model

> **Setup.** Agents of two types are arranged on a grid (or network). Each agent i has a threshold $\tau \in [0, 1]$ — the minimum fraction of same-type neighbors it needs to be "satisfied".
>
> **Update rule.**
>
> 1. Identify all *unsatisfied* agents (fewer than fraction τ of neighbors are same-type).
> 2. Each unsatisfied agent moves to a random vacant cell (or rewires a random tie in a network setting).
> 3. Repeat until no agent wants to move — or until a stable oscillation is reached.

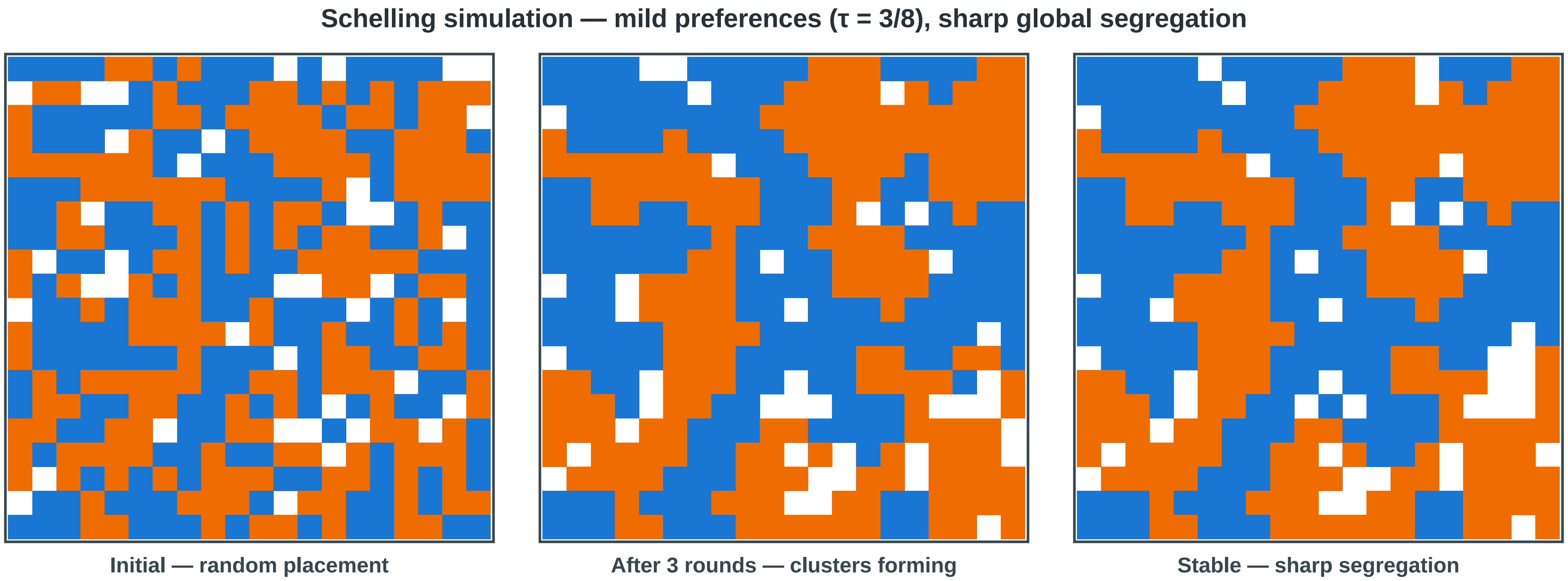
Schelling's insight (1971). Even τ as low as $1/3$ — "I just want a third of my neighbors to be like me" — produces **sharply segregated** global patterns.

One Step of the Model



The blue agent at (1,1) has 1/8 same-type neighbors — below $\tau = 3/8$. It relocates to vacant cell (4,3), where 3/5 neighbors are same-type — satisfied. Repeat across all unsatisfied agents until equilibrium.

Schelling Simulation



Source: Own Schelling simulation, in the spirit of Easley & Kleinberg (2010), Fig. 4.19

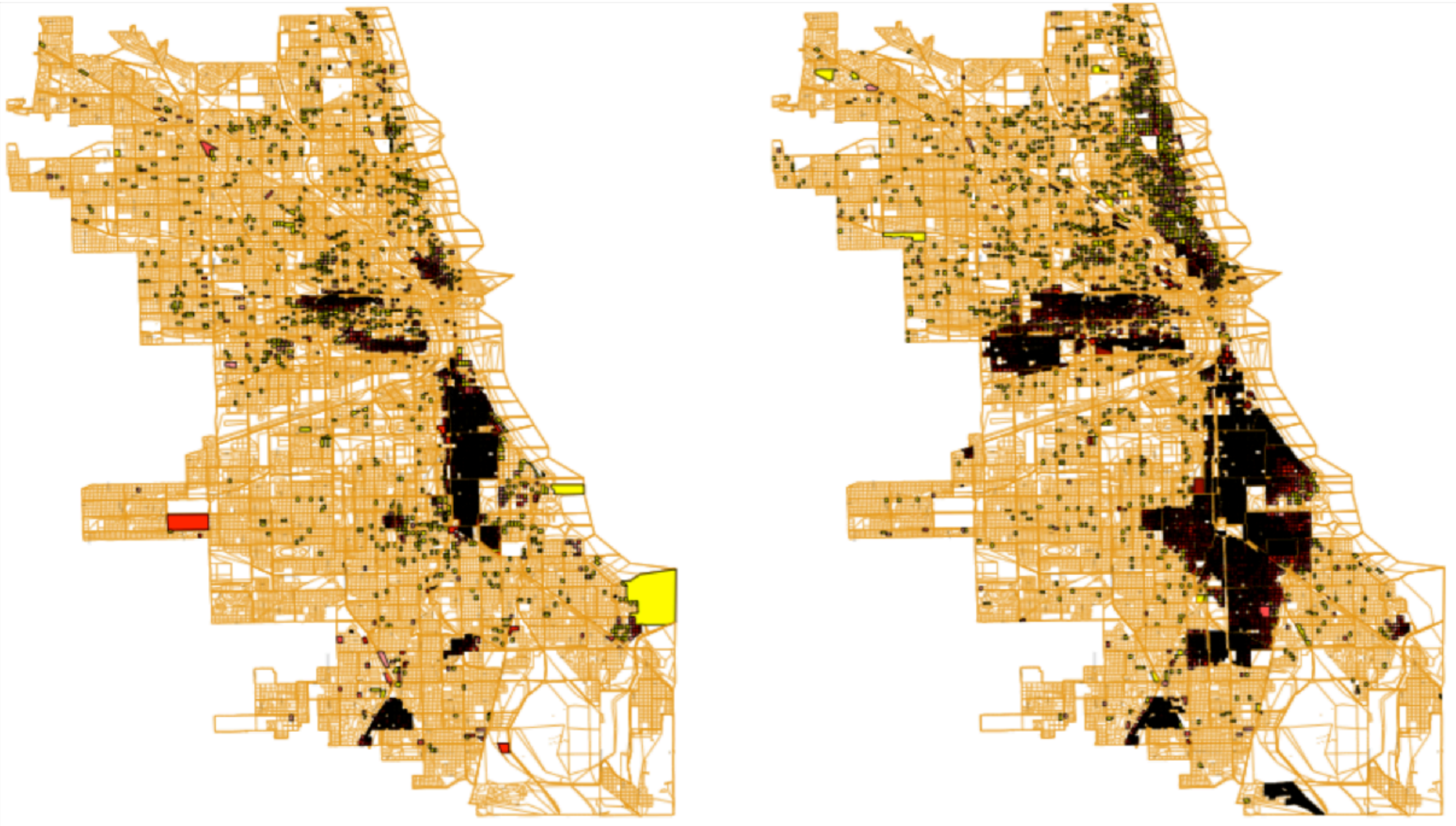
A 20×20 grid with $\approx!45$ blue, $\approx!45$ orange, $\approx!10$ vacant, run with threshold $\tau = 3/8$. Each round, every unsatisfied agent moves to a vacant cell where it *would* be satisfied; the process halts when nobody wants to move.

Within a handful of rounds, the locally-mild rule produces **macroscopically sharp** boundaries — exactly the picture the empirical map shows next.

Speaker notes

This is the simulation Easley & Kleinberg use (Networks, Crowds, and Markets, Fig. 4.19) to make Schelling's central point concrete. The preference is mild — $\tau = 3/8$ means "I just want a third of my neighbors to be like me" — yet at equilibrium each agent's neighborhood is overwhelmingly same-type. The macro-segregation is **dramatically disproportionate** to the micro-preference. Two lessons worth flagging: (1) global segregation does not require globally segregationist preferences; (2) it is the **iteration** of local moves that amplifies the mild rule — a one-shot placement under the same threshold would still look random. The next slide shows that real cities exhibit the same kind of sharp boundary the simulation produces.

Empirical Motivation



(a) *Chicago, 1940* (b) *Chicago, 1960*

Source: Easley & Kleinberg 2010

> Residential segregation in Chicago by race/ethnicity. The sharp boundaries between neighborhoods are consistent with Schelling dynamics: each household's mild preference for "some" same-type neighbors, iterated over decades of moves, produces stark global segregation.

Speaker notes

The empirical map raises the puzzle: how can boundaries be *this* sharp if individual preferences are not extreme? Schelling's model answers: iterated local moves amplify mild preferences into sharp global patterns. The same logic applies to online homophily, recommendation systems, and network rewiring.

From Grids to Networks

In a network setting, Schelling's model becomes a **rewiring** process:

- "neighbors" are graph-adjacent nodes
- "moving" becomes dropping a tie to a dissimilar contact and forming a new tie to a similar contact
- the threshold τ controls how much same-type concentration triggers rewiring

Speaker notes

This is the network-science translation of Schelling. The move from a spatial grid to a graph model changes the topology but preserves the core insight: mild local rules plus iterated updates produce sharp global segregation. The recommendation-algorithm connection brings the 1971 model into the 2020s: algorithms now run the Schelling update at machine speed, with consequences for filter bubbles, echo chambers, and political polarization.

From Grids to Networks

In a network setting, Schelling's model becomes a **rewiring** process:

- "neighbors" are graph-adjacent nodes
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- the threshold τ controls how much same-type concentration triggers rewiring

Implication for recommendation algorithms. If a platform suggests contacts who are *similar* to your current contacts, it acts as an automated Schelling rewirer — accelerating homophily-driven segregation even beyond what individual preferences would produce alone.

Quick Check

Translate Schelling's spatial model into a network setting.

1. What plays the role of "neighbor" in a network rather than a grid?
2. What plays the role of "moving"?
3. A social media platform recommends friends similar to your existing friends. In Schelling terms, what is the platform doing — and would you expect it to amplify or weaken homophily?

Speaker notes

Give them 4 minutes before revealing the fragments. (1) Network neighbors are graph-adjacent nodes. (2) "Moving" becomes rewiring — dropping ties to dissimilar contacts and adding ties to similar ones. (3) The platform acts as a Schelling updater: it identifies "unsatisfied" tie configurations (low similarity) and offers replacements (higher similarity). It amplifies homophily by accelerating the rewiring process.

Quick Check

Translate Schelling's spatial model into a network setting.

1. What plays the role of "neighbor" in a network rather than a grid?
2. What plays the role of "moving"?
3. A social media platform recommends friends similar to your existing friends. In Schelling terms, what is the platform doing — and would you expect it to amplify or weaken homophily?

1. Network neighbor: a graph-adjacent node — your immediate contacts.

2. Moving: rewiring — dropping ties to dissimilar contacts and forming new ones with similar contacts.

3. The platform is an automated Schelling rewirer. It surfaces similar people — equivalent to offering each agent a move toward a higher same-type fraction. This *amplifies* homophily: the local rule ("recommend similar") is mild, but at machine speed it produces sharp global outcomes (filter bubbles, echo chambers).

Takeaway

> Global polarization can emerge without anyone explicitly aiming for it. Mild local preferences plus iterated local updates produce sharp large-scale structure — and recommendation algorithms now run this process at scale.

Wrap-Up — Lecture 05

L05 added **context and time** to the graph. Nodes now have attributes, ties have timestamps, and social settings become explicit foci.

Question	L05 answer
Is similarity more than chance?	Compare observed homophily to a population baseline.
Why are similar people connected?	Distinguish selection, socialization, and shared context.
Where do ties form?	Model foci with affiliation networks and co-occurrence graphs.
How do local rules scale up?	Use closure curves and Schelling dynamics to connect micro behavior to macro patterns.

The central lesson: a graph snapshot can show a pattern, but explaining the mechanism requires baselines, temporal data, context, or a generative model.

Reading

Chapter 4 of Easley & Kleinberg (2010) covers homophily, selection, and socialization. McPherson, Smith-Lovin & Cook (2001), *Birds of a Feather*, is the definitive review of homophily research. Christakis & Fowler (2007) is the Framingham obesity study; Lyons (2011) provides the methodological critique.

Speaker notes

Close by emphasizing the L05 shift: the graph is no longer just topology. Attributes, timing, and social context determine what the observed pattern can mean. Homophily is measurable, but not self-explanatory; selection, socialization, shared foci, closure processes, and Schelling dynamics are different mechanisms that can produce similar-looking structure.

Image Credits & References

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